Optional Aesthetics

Each geometry type (points, lines, etc.) has a set of aesthetics that explain how visual elements relate to the data in our tabular dataset.

There are actually two different types of aesthetics: **required** and **optional**. We have to specify the required aesthetics. The optional ones have default values that we can use if we don't specify them in our code.

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For example, for geom_point we have:

Required	**x** and **y**
Optional	**color**, **size**, **shape**, **alpha**, **…**

Optional Aesthetics

Setting optional aesthetics works exactly like the required aesthetics, the only different being that we could leave them out if we wanted.

```
(ggplot(food) +
   geom_point(aes(x = "calories", y = "total_fat", color = "food_group")))
```

Fixed Aesthetics

Also, for any aesthetic we can specify a specific **fixed value** for all of the data in place of describing a column in the dataset. This is what is done internally for all of the optional aesthetics. For example, in geom_point color is set to "black" and size is set to 1.

The code for this requires a slightly different synax. The fixed aesthetic goes outside of the aes function:

```
(ggplot(food) +
   geom_point(aes(x = "calories", y = "total_fat"), color = "green"))
```

Note that when setting a numeric aesthetic such as size, we need to use the number without quotes around it.

Scales

One great thing about the grammar of graphics is that it allows us to stay in the abstract level where we can define the correspondence between elements of the plot and variables in our dataset. The software handles thigns such as the assigning of which color is associated with each category and where the axes should start and where the labels should go on them.

The element of the grammar of graphics that handles the translation from the data to the actual visual elements are called **scales**. Default scales exist for every geometry and will be used out-of-the-box. We can customize the scales used in the plot by selecting one and adding it inside of the ggplot code block.

Scales

Here is an example of changing the color scale used in one of the plots from the notes:

```
(ggplot(cbsa) +
   geom_point(aes(x='density', y='rent_1br_median', color='quad')) +
   scale_color_cmap_d())
```

It’s a common error to think that we add a scale to incorporate color into the plot. That’s not how it works! Color (and all aesthetics) get added within the geometry object. Scales only modify the aesthetics that we have already added above.

Advice for Graphics

As we continue working with data graphics this semester, keep the following things in mind. They will really help avoid errors and frustration!

– **Formatting:** Always put each outer function (ggplot, geom_, or scale_) on its own line and indent all but the first line by four spaces.
– **Copying Code:** Copy code from the notes and solutions is a good approach, particularly when you are just getting started or learning something new. However, I strongly recommend copying single ggplot, geom_ and scale_ functions rather than the entire block unless you really want to replicate a larger plot almost as-is.
– **Working through Notebooks**: It's always better to copy and edit code that you have already written in the notebook than going back to the notes. You'll already have made many of the changes that you need for the specific dataset we are working on.
– **When you are stuck**: If you get yourself really stuck on a question (in addition to asking for help with is always okay!), try to erase your block of code and start from scratchf.